Chapter 4 Additional and Advanced Exercises

8. An inequality

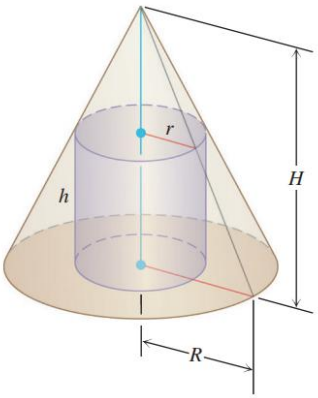
a. Show that $-1/2 \le x/(1 + x^2) \le 1/2$ for every value of x.

b. Suppose that f is a function whose derivative is $f'(x) = x/(1 + x^2)$. Use the result in part (a) to show that

$$|f(b) - f(a)| \le \frac{1}{2}|b - a|$$

for any a and b.

17. A max-min problem with a variable answer Sometimes the solution of a max-min problem depends on the proportions of the shapes involved. As a case in point, suppose that a right circular cylinder of radius r and height h is inscribed in a right circular cone of radius R and height H, as shown here. Find the value of r (in terms of R and H) that maximizes the total surface area of the cylinder (including top and bottom). As you will see, the solution depends on whether $H \le 2R$ or $H > 2R$.



21. To find $x = \sqrt[q]{a}$, we apply Newton's method to $f(x) = x^q - a$. Here we assume that a is a positive real number and q is a positive integer. Show that x_1 is a "weighted average" of x_0 and a/x_0^{q-1}, and find the coefficients m_0, m_1 such that

$$x_1 = m_0 x_0 + m_1\left(\frac{a}{x_0^{q-1}}\right), \quad \begin{array}{l} m_0 > 0, m_1 > 0, \\ m_0 + m_1 = 1. \end{array}$$

What conclusion would you reach if x_0 and a/x_0^{q-1} were equal? What would be the value of x_1 in that case?

24. Let $f(x)$, $g(x)$ be two continuously differentiable functions satisfying the relationships $f'(x) = g(x)$ and $f''(x) = -f(x)$. Let $h(x) = f^2(x) + g^2(x)$. If $h(0) = 5$, find $h(10)$.

30. Schwarz's inequality

a. In Exercise 29, let

$$f(x) = (a_1x + b_1)^2 + (a_2x + b_2)^2 + \cdots + (a_nx + b_n)^2,$$

and deduce Schwarz's inequality:

$$(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \le \left(a_1^2 + a_2^2 + \cdots + a_n^2\right)\left(b_1^2 + b_2^2 + \cdots + b_n^2\right).$$

b. Show that equality holds in Schwarz's inequality only if there exists a real number x that makes a_ix equal $-b_i$ for every value of i from 1 to n.

Exercises 5.2

Limits of Riemann Sums

For the functions in Exercises 39–46, find a formula for the Riemann sum obtained by dividing the interval $[a, b]$ into n equal subintervals and using the right-hand endpoint for each c_k. Then take a limit of these sums as $n \to \infty$ to calculate the area under the curve over $[a, b]$.

43. $f(x) = x + x^2$ over the interval $[0, 1]$.

45. $f(x) = 2x^3$ over the interval $[0, 1]$.

Exercises 5.3

Interpreting Limits of Sums as Integrals

Express the limits in Exercises 1–8 as definite integrals.

3. $\lim_{\|P\|\to 0} \sum_{k=1}^{n} (c_k{}^2 - 3c_k)\,\Delta x_k$, where P is a partition of $[-7, 5]$

5. $\lim_{\|P\|\to 0} \sum_{k=1}^{n} \frac{1}{1 - c_k}\,\Delta x_k$, where P is a partition of $[2, 3]$

Definite Integrals as Limits of Sums

Use the method of Example 4a or Equation (1) to evaluate the definite integrals in Exercises 63–70.

69. $\int_a^b x^3\,dx, \quad a < b$

71. What values of a and b maximize the value of

$$\int_a^b (x - x^2)\,dx?$$

(*Hint:* Where is the integrand positive?)

73. Use the Max-Min Inequality to find upper and lower bounds for the value of

$$\int_0^1 \frac{1}{1+x^2}dx.$$

85. Use the formula

$$\sin h + \sin 2h + \sin 3h + \cdots + \sin mh = \frac{\cos(h/2) - \cos((m + (1/2))h)}{2\sin(h/2)}$$

to find the area under the curve $y = \sin x$ from $x = 0$ to $x = \pi/2$ in two steps:

a. Partition the interval $[0, \pi/2]$ into n subintervals of equal length and calculate the corresponding upper sum U; then

b. Find the limit of U as $n \to \infty$ and $\Delta x = (b - a)/n \to 0$.

86. Suppose that f is continuous and nonnegative over $[a, b]$, as in the accompanying figure. By inserting points

$$x_1, x_2, \ldots, x_{k-1}, x_k, \ldots, x_{n-1}$$

as shown, divide $[a, b]$ into n subintervals of lengths $\Delta x_1 = x_1 - a$, $\Delta x_2 = x_2 - x_1, \ldots, \Delta x_n = b - x_{n-1}$, which need not be equal.

a. If $m_k = \min\{f(x) \text{ for } x \text{ in the } k\text{th subinterval}\}$, explain the connection between the **lower sum**

$$L = m_1 \Delta x_1 + m_2 \Delta x_2 + \cdots + m_n \Delta x_n$$

and the shaded regions in the first part of the figure.

b. If $M_k = \max\{f(x) \text{ for } x \text{ in the } k\text{th subinterval}\}$, explain the connection between the **upper sum**

$$U = M_1 \Delta x_1 + M_2 \Delta x_2 + \cdots + M_n \Delta x_n$$

and the shaded regions in the second part of the figure.

c. Explain the connection between $U - L$ and the shaded regions along the curve in the third part of the figure.

87. We say f is **uniformly continuous** on $[a, b]$ if given any $\epsilon > 0$, there is a $\delta > 0$ such that if x_1, x_2 are in $[a, b]$ and $|x_1 - x_2| < \delta$, then $|f(x_1) - f(x_2)| < \epsilon$. It can be shown that a continuous function on $[a, b]$ is uniformly continuous. Use this and the figure for Exercise 86 to show that if f is continuous and $\epsilon > 0$ is given, it is possible to make $U - L \leq \epsilon \cdot (b - a)$ by making the largest of the Δx_k's sufficiently small.

Exercises 5.4

Evaluating Integrals

Evaluate the integrals in Exercises 1–28.

14. $\displaystyle\int_{-\pi/3}^{\pi/3} \sin^2 t \, dt$

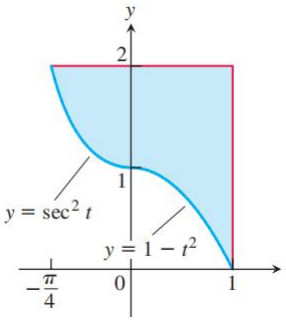
28. $\displaystyle\int_0^{\pi} \frac{1}{2}(\cos x + |\cos x|) \, dx$

Find dy/dx in Exercises 39–46.

42. $y = x \displaystyle\int_2^{x^2} \sin(t^3) \, dt$

Find the areas of the shaded regions in Exercises 51–54.

54.



69. Find the linearization of

$$f(x) = 2 - \int_2^{x+1} \frac{9}{1+t}\,dt$$

at $x = 1$.

72. Another proof of the Evaluation Theorem

a. Let $a = x_0 < x_1 < x_2 \cdots < x_n = b$ be any partition of $[a, b]$, and let F be any antiderivative of f. Show that

$$F(b) - F(a) = \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})].$$

b. Apply the Mean Value Theorem to each term to show that $F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1})$ for some c_i in the interval (x_{i-1}, x_i). Then show that $F(b) - F(a)$ is a Riemann sum for f on $[a, b]$.

c. From part (b) and the definition of the definite integral, show that

$$F(b) - F(a) = \int_a^b f(x)\,dx.$$

Additional exercises.

*Ex.1.** Assume $f(x)$ and $g(x)$ are integrable over a bounded interval $[a,b]$. Show that $f(x)+g(x)$ and $f(x)g(x)$ are also integrable over $[a,b]$.

*Ex.2.** Examine integrability of the following functions over $I=[0,1]$

(a) $f(x)=\begin{cases}1-\frac{1}{n}, & x\in(\frac{1}{2^n},\frac{1}{2^{n-1}}], \quad n=1,2,\cdots,\\ 1, & x=0.\end{cases}$

(b) $f(x)=\begin{cases}x, & x \text{ is rational},\\ 2x, & x \text{ is irrational}.\end{cases}$

(c) $f(x)=u'(x), \quad u(x)=\begin{cases}x^2\sin\frac{1}{x}, & x\neq 0,\\ 0, & x=0.\end{cases}$

(d) $f(x)=v'(x), \quad v(x)=\begin{cases}x^2\sin\frac{1}{x^2}, & x\neq 0,\\ 0, & x=0.\end{cases}$

Ex.3. (a) Assume g is continuous on $[a,b]$ and f is continuous on the range of g. Show that $f \circ g$ is continuous on $[a,b]$, hence is integrable over $[a,b]$.

(b) Assume f is integrable over $[a,b]$ and $f \circ f$ is well-defined on $[a,b]$. Must $f \circ f$ be integrable over $[a,b]$?

More general, assume f and g are integrable over $[a,b]$ and $f \circ g$ is well-defined on $[a,b]$. Must $f \circ g$ be integrable over $[a,b]$?

(c) Assume f is bounded but not integrable over $[a,b]$. Assume $f \circ f$ is well-defined on $[a,b]$. Must $f \circ f$ be non-integrable over $[a,b]$?

More general, assume f and g are bounded but not integrable over $[a,b]$. Assume $f \circ g$ is well-defined on $[a,b]$. Must $f \circ g$ be non-integrable over $[a,b]$?

*Ex.4.** Assume $F(x)$ is continuous on $[a,b]$ and $F'(x) = f(x)$ for all x in (a,b) except for a finite number of points, and f is integrable on $[a,b]$. Show that

$$\int_a^b f(x)dx = F(b) - F(a).$$

*Ex.5.** Find

$$\frac{d}{dx}\int_0^x f(t)dt$$

for the given f.

(a) $f(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases}$

(b) $f(x) = \begin{cases} 1, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$